# Functions & Libraries

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In the last part I said **we want to take that code we created**, and make it into something called a function. Why we do this is so we can; Easily add to our program, have it look better, *or not have to rewrite code again if we want to do something similar.*

Let's turn the code we had to calculate the price of the shopping cart, **into a function.**

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**First,** before we need to understand what a function is again.

- Remember in math with y = x, or f(x) = x. Notice how you need to put something in to get a result, **it needs something to do it's job as a function.** That's called an argument, x is an argument.

```
person = {"name": "Bob", "balance": 100, "discount": 0.1, "cart": [{ "strawberry": 2 }, { "banana": 3 }]}
```

**In** our code we have from the previous part. **WE USED** `person["cart"]` to complete our instructions (**Go back and look at, *what we needed* to complete our code**).

- That means we NEED to have it in a function so an ARGUMENT.

> *An important thing is, when changing something into function.* **The code isn't really going to change at all,** we're using the same code.

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```
def calculate_total(cart):
```

- This is how you **create a function,** you say *def* to "define" THEN, you say some word that describes what you want that action to do. **Then give another describing placeholder word for the argument.**
  - **cart,** is a supposed to be a placeholder to equal what you want to give it. We want to give it `person["cart"]`. **cart is the SAME as person["cart"].**

> The last important things about functions, is that they have something called a **RETURN VALUE,** meaning this function EQUALS this value.

- **All** that really is, whatever variable that you calculated you say `return variable`. **This is just the output of the function.**

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```
def calculate_cart_total(cart):
    total = 0
    for item in cart:
        for price in item.values():
            total += price
    return total
```

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This is the ENTIRETY of our function, the only thing we changed *(If you look back to the code in the last part.)* was **person[“cart”]** became **cart, and we added something called a return value.**

- OK, let’s actually use this.

So now, whenever we want to calculate the total of the cart, all we have to do, is give this **FUNCTION** the person’s cart.

```
calculate_cart_total(person["cart"])
```

- **This is equal to the total of the cart,** it doesn’t print anything. To print it, just print it.

Now we can **more cleanly ORGANIZE our code,** let’s say we have an if statement as a *“customer interaction area”*. that allows you do say 1,2,3,4,5 for the options you want to chose.

- **INPUTTING 1** let’s you see everything in the product catalogue.
- **INPUTTING 2** let’s you add stuff to your cart.
- **INPUTTING 3** let’s you see your cart total
- **INPUTTING 4** let’s you pay.

If we DIDN’T have functions, we’d have to put that entire for loop inside that **IF STATEMENT.** Instead, we can just use ONE line to cleanly create this *“customer interaction area”* for each option.

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Let’s quickly do it again for the “**VIEW CART**”

```
def view_cart(cart):
    for item in cart:
        print(item)
```

Notice, again all we changed was cart is now replacing **person[“cart”]**

- **Again,** we use this function by doing `view_cart(person["cart"])`

We don't always have to **MAKE our own functions,** Libraries are community functions we can add and use.

- Let's give EACH person an ID, we could just do 1,2,3,4 but that's not safe, we could use random numbers; that would be using a library. The issue with that is `random` **numbers may product duplicates**

*Luckily*, there is a library called **UUID,** which is used to create ID's that almost never makes duplicates.

To get a library we do

```
import library
```

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The library we're grabbing from is called UUID, **the FUNCTION is called UUID4.**

```
import uuid
```

SAVE the **variable, ADD** it to the person object.

- For functions we CALL them, *that is done by doing* object.functionname()

```
user_uuid = uuid.uuid4()
```

**ADD,** the ID to our object

```
person = {"id": user_uuid, "name": "Bob", "balance": 100, "discount": 0.1, "cart": [{ "strawberry": 2 }, { "banana": 3 }]}
```

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That's all for functions, the next part will cover **OBJECTED ORIENTED PROGRAMMING,** which sounds fancy and complicated. But it's just combining *functions and objects*.